

You may soon be able to stretch your TV



Stanford University researchers have developed a proof-of-concept of a reshapable display made up of a stretchy polymer called 'super yellow' and the synthetic plastic material polyurethane. It has a maximum brightness of at least two times that of a cellphone screen and can be stretched up to twice its original length without tearing. The display can be stuck to your skin, making wearable electronics a valuable market for this technology. "Imagine a display where you can both see and feel the three-dimensional object on the screen," says Stanford University chemical engineer Zhenan Bao.

Zhitao Zhang wears the flexible light-emitting film, featuring a Stanford logo, on the knuckle of their finger. This shows how the film can hold up to flexing and wrinkling (Credit: Zhitao Zhang and Jiancheng Lai of Bao Group Research Lab)

The lifespan of lithium-ion batteries doubled

Researchers at the University of Queensland have developed a technology that more than doubles the lifespan of highly sought-after high-voltage lithium-ion batteries, which achieve higher energy density but last only several hundred cycles. The researchers have demonstrated a battery that remains stable for more than 1000 charge/discharge cycles. They designed a uniquely-grown atomic-thin functional layer on the surface of a high-voltage cathode, which is the source of lithium ions and is a critical aspect that limits the cycle life in a battery. It should be noted that with the industry under increasing pressure to decarbonise, the development of lithium-ion batteries of lower cost, higher energy density, and longer cycle life is vitally important.

Atomic-thin layer grown on cathode enables long-lasting battery (Credit: <https://www.uq.edu.au>)

